Question	Answer	Marks	Partial Marks
1(a)(i)	3050	**2**	**M1** for 50 × 40 + 70 × 15 or better
1(a)(ii)	91.1 or 91.14 to 91.15	**1**	**FT** $\frac{2780}{their\,3050}\times 100$
1(a)(iii)	125 nfww	**2**	**M1** for [...] × $\frac{100-4}{100}$ =120 oe
1(a)(iv)(a)	132	**2**	**B1** for increase of 6 in adult or junior or **M1** for 56 : 76 or for multiples of 33 seen 33, 66, 99, 132, … or 50 + x : 70 + x = 14 : 19 oe or (70 – 50) × $\frac{19+14}{19-14}$ oe or 50 + x = (120 + 2x) × $\frac{14}{19+14}$ oe
1(a)(iv)(b)	10	**2**	**FT** $\frac{their\,\mathbf{(a)}-120}{120}\times 100$ dep on *their* **(a)** > 120 **M1** for $\frac{their\,\mathbf{(a)}-120}{120}[\times 100]$ or $\frac{their\,\mathbf{(a)}}{120}\times 100\,[-100]$
1(b)(i)	2280 or 2281 to 2282 nfww	**2**	**M1** for $2500\times\left(1-\frac{3}{100}\right)^3$ oe
1(b)(ii)	8	**2**	**M1** for $2500\times\left(1-\frac{3}{100}\right)^n$ or 0.97^n evaluated with $n > 3$
2(a)(i)	56	**1**	
2(a)(ii)	11	**2**	**M1** for $120 - n^3 = -1211$ or $120 - 11^3 = -1211$
2(b)	0.0048 or $\frac{3}{625}$ oe	**1**	

Question	Answer	Marks	Partial Marks
2(c)	A: –5, $10 - 3n$ B: $\frac{5}{8}$, $\frac{n}{n+3}$ C: 20, $n^2 - n$	**8**	**B1** for –5 **B2** for $10 - 3n$ oe or **B1** for $k - 3n$ or for $10 - kn$ **B1** for $\frac{5}{8}$ **B1** for $\frac{n}{n+3}$ oe **B1** for 20 **B2** for $n^2 - n$ oe or **B1** for any quadratic or for at least two second differences of 2
3(a)	5	**B1**	
	4	**B1**	
	3.55	**3**	**M2** for (10 × 1 + 6 × 2 + 11 × 3 + 13 × 4 + 14 × 5 + 6 × 6) ÷ 60 oe or **M1** for 10 × 1 + 6 × 2 + 11 × 3 + 13 × 4 + 14 × 5 + 6 × 6 oe
3(b)(i)	42.55 or 42.6	**4**	**M1** for 25, 40, 62.5 soi **M1** for Σfx with x values in correct intervals, including boundaries **M1 dep** on second M1 for $\frac{\Sigma fx}{100}$
3(b)(ii)	10.8 2.16	**2**	**B1** for each or for frequency densities 3.6 and 0.72 seen
4(a)	$\frac{1}{2}(r+5)(r+2)\sin 30 = (r+1)^2$	**M2**	**M1** for $\frac{1}{2}(r+5)(r+2)\sin 30$ oe
	$r^2 + 5r + 2r + 10$ or $r^2 + r + r + 1$ soi	**B1**	
	Leading to $3r^2 + r - 6 = 0$ with no errors or omissions	**A1**	Dependent on both expansions seen